Institutional Integration of 10Hz NFL Spatial Tracking into the Beexly/Sports Probability Engine: Geometric Deep Learning, Continuous-Time Valuation, and Fail-Closed Conformal Architectures
Geometric Deep Learning and Permutation-Equivariant Spatio-Temporal Architectures
Transitioning a predictive probability engine from play-by-play discrete event logs to continuous micro-movement tracking requires reformulating the state representation of American football. In the National Football League's Next Gen Stats (NGS) platform, player tracking data is captured via ultra-wideband radio-frequency identification (RFID) tags embedded in player shoulder pads and footballs at a frequency of 10 Hz. Each raw frame corresponds to a 100-millisecond snapshot capturing two-dimensional Cartesian coordinates (x,y), speed s, acceleration a, distance traversed Δd, orientation angle θ
o

 , and velocity motion angle θ
dir

 .

Standard coordinate spaces span a 120-yard longitudinal axis (including 10-yard end zones) and a 53.3-yard lateral axis. Prior to model ingestion, all plays must be standardized to a canonical frame of reference where the offensive team moves along the positive x-axis from left to right. For a play spanning T time frames with N=22 players and the ball (N
a

 =23), the spatio-temporal state tensor is denoted as Z∈R
T×N
a

 ×D
 , where feature dimension D=10 encapsulates:

z
i,t

 =[x
i,t

 ,y
i,t

 ,v
x,i,t

 ,v
y,i,t

,a
x,i,t

,a
y,i,t

,sin(θ
o,i,t

),cos(θ
o,i,t

),sin(θ
dir,i,t

),cos(θ
dir,i,t

)]
T

A persistent challenge in deep modeling of team invasion sports is permutation equivariance. Arbitrary indexing of offensive skill players or defensive backs introduces an artificial factorial ordering complexity (11!≈3.99×10
7
). If an architecture is sensitive to index permutations within player groups, sample efficiency degrades, and generalization on unseen roster permutations collapses.

Early spatio-temporal graph neural networks (ST-GNNs) modeled players as graph nodes connected by static or dynamic adjacency edges, executing spatial graph convolutions interleaved with temporal long short-term memory (LSTM) or temporal convolutional networks (TCNs). However, message passing across fixed graph topologies fails to capture long-range multi-body field dynamics, such as a backside safety reading an offset boundary tight end.

Modern trajectory transformer architectures resolve this through Set Attention Blocks (SABs) that operate symmetrically over sets of agents while tracking temporal evolution. For a set of player representations H
t

∈R
N×d
, an agent-permutation equivariant set attention layer is defined via multi-head attention:

MHA(Q,K,V)=Softmax(
d
k

QK
T

+B
rel

)V
where Q=H
t

W
Q

, K=H
t

W
K

, and V=H
t

W
V

. The structural spatial geometry is injected directly through a continuous spatial bias tensor B
rel

∈R
N×N
, which encodes pairwise relative positions and Euclidean distances between player i and player j:

B
rel

(i,j)=ϕ(p
i,t

−p
j,t

, // p
i,t

−p
j,t

//
2

)
where ϕ(⋅) is a multi-layer perceptron projecting Euclidean spatial offsets into attention logit adjustments.

Simultaneous full attention across both the agent axis (N=22) and the temporal sequence (T≈50–100 frames) incurs a quadratic complexity of O((NT)
2
), which is computationally prohibitive for millisecond-latency streaming inference. The engine utilizes factorized spatio-temporal attention. The architecture decouples the transformation into sequential stages:

First, a temporal self-attention block models the trajectory history of each individual player independently across the sequence length T, scaling as O(NT
2
). Second, an equivariant spatial cross-attention block computes multi-agent interaction and defensive gravity across the N agents at each discrete time step t, scaling as O(TN
2
). This factorized design reduces the total attention complexity to O(NT
2
+TN
2
), decreasing floating-point operations by over 80% while preserving physical plausibility and temporal consistency.

For play forecasting and counterfactual simulation, the system integrates a non-autoregressive parallel prediction architecture modeled on Mixture-of-Gaussians trajectory predictors (e.g., PlayGen-MoG). Rather than predicting sequential displacements autoregressively—which

accumulates compounding drift over multi-second horizons—the network conditions on the static pre-snap formation $F \in R^{N \times 2}$ and role identifiers $r \in \{0,\ldots,7\}^{N}$ to output parameters for K discrete multimodal Gaussian mixture components simultaneously across all T timesteps:

$$P(X \mid F,r) = \sum_{k=1}^{K} \pi_k \prod_{t=1}^{T} \prod_{i=1}^{N} N(x_{i,t}; \mu_{i,t}^{(k)}, \Sigma_{i,t}^{(k)})$$

Critically, the mixture weights π_k are shared across all 22 agents. This joint coupling ensures that the sampled mixture component reflects a globally coordinated tactical play scenario (e.g., a specific coverage rotation countering a specific pass concept) rather than independent, fragmented player

behaviors, structurally preventing mode collapse and maintaining physical coordination across all players.

Architectural Component	Formulation / Implementation	Invariance / Equivariance	Computational Complexity	Primary Operational Failure Mode
ST-GNN (Baseline)	Spatial Graph Conv + Temporal LSTM/GRU	Permutation-invariant via node pooling	$O(T \cdot \|E\| + T \cdot N \cdot d^2)$	Fails on long-range non-local coverage assignments.
Full Spatio-Temporal Transformer	Joint Self-Attention over flattened (N×T) tokens	Permutation-equivariant if role embeddings omitted	$O(N^2 T^2 \cdot d)$	Out-of-memory errors on GPU; exceeds 100ms latency envelope.
Factorized Spatio-Temporal Transformer	Sequential: Temporal Attention → Spatial Set Attention	Agent-permutation equivariant; strictly causal in time	$O(NT^2 \cdot d + TN^2 \cdot d)$ [cite: 7]	Sub-optimal when high-frequency acceleration alters spatial topology within <50ms.
Non-Autoregressive PlayGen-MoG	Relative Spatial Attention + Multi-Agent Mixture-of-Gaussians Head			

Permutation-equivariant with role-conditioned coordinate bias

$O(K \cdot T \cdot N^2 \cdot d)$ (single pass)

Component entropy collapse if dispersion priors are unconstrained.

Continuous-Time Within-Play Valuation and Fluid Dynamics Pitch Gravity
Traditional football analytics evaluates performance through discrete Expected Points Added (EPA) and Win Probability (WP) differentials mapped at the play-by-play boundary. These models obscure the intra-play micro-actions that generate value. To capture within-play value generation, the engine models a continuous-time Markov decision process (CTMDP) defined over the play duration $t \in [0,\tau]$, formalizing the within-play expected points value $V(s_t)$ as a continuous-time semi-martingale:

$$V(s_t)=E[EP_{t_{end}} \mid F_t]$$

where F_t denotes the continuous filtration generated by the tracking tensor up to time t, and $EP_{t_{end}}$ is the expected points at the terminal play state.

Following the framework established by Yurko et al. (2020) in Going Deep, continuous-time within-play valuation is decomposed into modular, conditional sub-models. At each frame t, the expected value of an ongoing play depends on the ball state, separating into pre-throw pass valuation, ball flight valuation, and ball-carrier valuation:

$$V(s_t) = I_{\text{pre-throw}} \cdot V_{\text{pass}}(s_t) + I_{\text{flight}} \cdot V_{\text{flight}}(s_t) + I_{\text{carrier}} \cdot V_{\text{run}}(s_t)$$

For a ball carrier (either a running back post-handoff or a receiver post-catch), the terminal yards gained Y is a continuous random variable conditioned on the instantaneous spatial configuration of the 11 defensive players and 10 blockers. Rather than estimating a deterministic expectation

$$\hat{Y} = E[Y \mid s_t]$$

], which obscures bimodal outcomes such as goal-line stands versus breakout touchdowns, the engine deploys flexible random forests and conditional normalizing flows for continuous conditional density estimation:

f
Y∣S

(y∣s
t

)=P(Y=y∣p
carrier

,v
carrier

,{p
def
j

,v
def
j

}
j=1
11

,{p
off
k

,v
off
k

}
k=1
10

)
The instantaneous expected value is computed by integrating the continuous Expected Points curve g
EP

(·) over the estimated density:

V
run

(s
t

)=∫
−∞
∞

g
EP

(x
LOS

+y)·f
Y|S

(y|s
t

)dy
To evaluate movement efficiency, the engine compares the observed ball carrier trajectory against a posterior predictive distribution of counterfactual "ghost" defenders and alternative carrier steps. Following the Bayesian step-and-turn movement formulation (Nguyen & Yurko, 2026), player motion is decomposed into step length Δl
t

=∥p
t

$-p$
$t-1$

$//$
2

and turn angle $\Delta\theta$
t

$=\theta$
dir,t

$-\theta$
$dir,t-1$

. Step length is modeled via a Gamma likelihood with log-link, while turn angle is modeled via a circular von Mises distribution with concentration parameter κ
i

capturing individual directional agility:

$\Delta\theta$
i,t

$\sim von\ Mises(\mu$
i,t

$,\kappa$
i

$),f(\theta;\mu,\kappa)=$
$2\pi I$
0

(κ)
$\exp(\kappa\cos(\theta-\mu))$

Integrating high-frequency player velocities across space establishes an empirical vector field over the gridiron, adapting continuous fluid dynamics principles. Discretizing the field into a mesoscale grid of cells Ω
m,n

of dimension 0.5×0.5 yards, the accumulated player velocity vector field

u

(x,t) is reconstructed from player trajectory crossings:

u

(x,t)=
∑
i=1
N

w
i

(t)K(x−p
i,t

)
∑
i=1
N

w
i

(t)K(x−p
i,t

)
v

i,t

where K(⋅) is an anisotropic spatial Gaussian kernel and w
i

(t) represents the positional influence weight. Under the Helmholtz-Hodge decomposition, any sufficiently smooth vector field decomposes into irrotational (curl-free) and solenoidal (divergence-free) components:

u

(x)=−∇Φ(x)+∇×A(x)
Empirical evaluation of gridiron tracking data demonstrates that collective team movements are predominantly irrotational (∇×
u

≈0), as players move directionally toward targeted spatial waypoints (e.g., pursuit angles, route stems) governed by tactical constraints. Consequently, the vector field is generated by a latent scalar potential Φ(x), satisfying Poisson's equation:

∇
2
Φ(x)=−ρ
gravity

(x)
where ρ
gravity

(x)=∑
j∈Def

q
j
def

δ(x−p
j

)−∑
k∈Off

q
k
off

δ(x−p
k

) models the spatial gravity charge of defensive and offensive agents.

The defensive scalar potential Φ
def

(x) represents the spatial friction or resistance of the gridiron terrain. The gradient vector field
E

gravity

$=-\nabla\Phi$
def

(x) indicates the directional force exerted by the defense. Running lanes and passing windows correspond directly to potential wells—regions where Φ
def

(x) is minimized and the local dynamic pressure gradient $\nabla P=\rho($
u

$\cdot\nabla)$
u

exhibits divergent expansion, signaling defensive structural breakdown.

Granular Route Geometry, Phase-Segmented Separation, and Leverage Tensors
Evaluating wide receiver performance via gross Average Separation at the catch point conflates receiver separation skill with defensive diagnostic capability and quarterback timing. A wide receiver running an uncontested drag route against a busted Cover-0 blitz exhibits massive catch-point separation despite zero route running win margin. To isolate intrinsic route-running talent, separation must be modeled continuously as a function of route geometry, defensive leverage, and coverage scheme baselines.

The raw 10Hz spatial coordinates of a receiver trajectory p
rec

$(t)=[x(t),y(t)]$
T
are parameterized into continuous, twice-differentiable curves using B-spline basis functions of degree p=3 across knot sequence τ
0

$\leq\tau$
1

$\leq\cdots\leq\tau$
m

:

p
rec

(t)=
i=0
∑
n

P
i

B
i,p

(t)
where P
i

are control points in R
2
and B
i,p

(t) are calculated recursively via the Cox-de Boor formula. Parameterizing the trajectory into continuous splines allows exact analytical extraction of the kinematic derivatives: instantaneous velocity
v

(t)=
p
˙

(t), acceleration
a

(t)=
p
¨

(t), and trajectory curvature κ(t):

$$\kappa(t)=\frac{(\dot{x}(t)^2+\dot{y}(t)^2)^{3/2}}{|\dot{x}(t)\ddot{y}(t)-\dot{y}(t)\ddot{x}(t)|}$$

The trajectory is segmented into three distinct kinematic phases:

The route progression originates at the line of scrimmage snap and proceeds through the stem phase ($t_0 \leq t < t_{break}$), characterized by vertical acceleration and leverage manipulation. The break point corresponds to the global maximum of trajectory curvature $\kappa(t)$ exceeding an angular threshold ($\kappa(t$

break

)>κ
threshold

), accompanied by a localized deceleration dip //
a

(t) // <0 followed by directional realignment. Finally, the target arrival phase (t
break

≤t≤t
target

) spans the post-break trajectory to the arrival of the football.

Receiver separation S(t)=min
j∈Def

// p
rec

(t)−p
def
j

(t) //
2

depends heavily on the defender's pre-break leverage tensor L
def

(t). Leverage is a multi-dimensional spatial construct capturing lateral shade, vertical cushion, and body orientation mismatch. Let d
ij

(t)=p
rec

(t)−p
def
j

 (t) represent the relative displacement vector. The defender leverage tensor L
ij

 (t)∈R
4
  is formalized as:

L
ij

 (t)=

d
x,ij

 (t)
d
y,ij

 (t)
cos(θ
o,def
j

 (t)−θ
dir,rec

 (t))
cos(θ
o,def
j

 (t)−θ
dir,def
j

(t))

The third component evaluates the defender's gaze and hip orientation relative to the receiver's running vector. The fourth component captures the defender's biomechanical transition state: when a cornerback backpedals, cos(θ
o

−θ
dir

)≈−1.0, indicating that their hips are opened toward the line of scrimmage while traveling backward.

When the receiver triggers their break, if the defender's hip orientation mismatch Δθ
hip

=∣θ
o,def

−θ
dir,rec

∣>90
∘
, the defender must execute a complete hip flip. This biomechanical constraint enforces a transition recovery delay of 200–400ms during which the defender's instantaneous acceleration drops to near zero, creating an immediate separation window.

To compute Expected Separation Over Expected (ESOE), the engine trains a gradient-boosted quantile regressor to model the conditional distribution of separation across the route progression:

E[S(t
arrival

)∣RouteClass,CoverageScheme,L(t
break

),
v

rec

(t
break

),t
throw

−t
break

]
The residual metric, ESOE=S(t
arrival

)−E[S(t
arrival

)], isolates pure receiver route-running precision by controlling for defensive cushion, coverage identity, and pass timing.

Route Kinematic Phase	Time Boundary	Kinematic Identifiers	Modeled Physical Mechanism
Phase 1: Stem Progression	t 0 ≤t<t break		

[cite: 12]

High linear acceleration (//
a

// >3.5 m/s
2

), low path curvature (κ(t)≈0).

Vertical push, attacking defender cushion, and forcing defensive hip commitments.

Phase 2: Break Point Execution
t=t
break

[cite: 12]

Curvature peak (κ(t)>κ
threshold

), localized deceleration dip (//
a

// <0).

Directional cut, center-of-mass lowering, and exploitation of defender leverage.

Phase 3: Target Arrival Window
t
break

<t≤t
target

[cite: 12]

Re-acceleration along break vector, ball flight convergence.

Separation maintenance, defensive hip-flip recovery exploitation, and catch-point framing.

Probabilistic Coverage Attribution and Safety Rotation Disguise Mechanics
Historical coverage metrics relied on nearest-defender attribution at the catch point, misattributing blame on approximately 25% of passing plays due to zone hand-offs, blown assignments, and safety rotations. The engine resolves this by deploying a factorized attention transformer that calculates frame-by-frame assignment probabilities for all 11 defensive players from pre-snap alignment through pass release.

Let $A=\{a_1, a_2, \ldots, a_{20}\}$ represent the set of 20 discrete defensive coverage assignments, encompassing specific zone responsibilities (Deep 1/2, Deep 1/3, Deep 1/4, Flat, Hook-Curl, Hole) and man-to-man coverage matchings. For each frame t, the transformer evaluates defensive spatial embeddings $h_i(t)$ and eligible offensive receiver embeddings $g_j(t)$.

The model outputs two coupled probability distributions: an assignment tensor $P_{assign} \in R^{11\times 20}$ where $P_{i,k}(t)=P(\text{Assignment}_i = a_k \mid F_t)$, and an attention-weighted bipartite matching matrix $M \in [0,1]^{11\times 5}$ where $M_{ij}(t)=P(\text{Def}$

i

↔Rec
j

|F
t

) captures the exact pairing probability between defender i and receiver j.

Defensive coverage disguise is quantified by tracking the dynamic divergence of coverage probabilities across the pre-to-post-snap boundary. Defenses show an initial pre-snap look C
pre

(e.g., a two-high safety shell indicating Cover-2 or Cover-4) and rotate at the snap into a single-high closed structure C
post

(Cover-1 or Cover-3). The disguise detection model defines the instantaneous Disguise Entropy H
disguise

(t) as the informational divergence between pre-snap alignment priors and post-snap trajectory kinematics:

D
disguise

=D
KL

(P
coverage

(t
post

)// P
coverage

(t
snap

))=

$$c \in C$$
$$\sum$$

$$P(c \mid F$$
$$t$$
$$post$$

$$)\ln($$
$$P(c \mid F$$
$$t$$
$$snap$$

$$)$$
$$P(c \mid F$$
$$t$$
$$post$$

$$)$$

$$)$$
Safety rotation dynamics serve as the primary operational tell. The engine monitors the kinematic centroid and differential velocity of the free safety (FS) and strong safety (SS). Let y
$$FS$$

$$(t),y$$
$$SS$$

$$(t)$$ represent lateral coordinates and v
$$x,FS$$

$$(t),v$$
$$x,SS$$

$$(t)$$ represent downhill longitudinal velocities. A safety rotation probability index R
$$safety$$

$$(t)$$ is parameterized as:

R
safety

(t)=σ(w
T
[Δy
safeties

(t),v
x,SS

(t)−v
x,FS

(t),depth
SS

(t)−depth
FS

(t),cos(θ
dir,SS

(t))]
T
)
When the strong safety triggers a downhill rotation into the intermediate hook-curl or flat zone while the free safety rotates toward the middle third of the field, R
safety

(t) surges above 0.85 within 400ms of the snap. This allows the engine to lock the true post-snap coverage identity long before quarterback pass release, establishing an information edge over betting markets that trade live derivative lines on delayed television broadcasts.

Defensive Coverage Scheme Pre-Snap Structural Tell Frame-by-Frame Transition Dynamics Post-Snap Terminal Assignment (t=trelease)
Cover-0 (All-Out Blitz)
Zero safeties deep (<12 yards depth); cornerbacks in press alignment (cushion<2 yards).

Immediate forward surge (
v

.
x

^
>3.0 m/s) by linebackers and safeties into pass rush.

Man-to-man across all eligible receivers; 0 deep zone help defenders.

Cover-1 (Man-Free)
Single-high safety positioned between hashes at 15–18 yards depth; boundary CBs with outside shade.

Free safety stabilizes in center field (//
v

// ≈0); strong safety or slot defender drops into "rat" hole or man coverage.

1 deep post-safety; 5 intermediate man-to-man assignments.

Cover-2 (Two-Deep Zone)
Two safeties split wide outside the hash marks at 12–15 yards depth.

Both safeties backpedal into deep halves (v
x

<−2.0 m/s); boundary CBs force inward release and hold flat zones.

2 deep halves; 5 underneath defenders in hook-curl and flat zones.

Cover-3 (Three-Deep Zone)
Often disguised as two-high shell; one safety exhibits creeping downhill motion pre-snap.

Downhill safety rotates rapidly into the flat/curl; remaining safety sprints laterally to deep middle third.

3 deep thirds (2 boundary CBs + 1 FS); 4 underneath zone defenders.

Cover-4 (Quarters Zone)
Two safeties at 10–12 yards depth aligned directly over interior receivers.

Safeties read releases of #2 receivers; if #2 goes vertical, safety matches man-to-man over the top.

4 deep quarters; 3 underneath zone defenders.

Cover-6 (Quarter-Quarter-Half)
Asymmetric safety alignment: boundary safety deeper than field safety.

Field side executes Cover-4 matching; boundary side executes Cover-2 flat rotation.

Split-field: 1 deep half boundary, 2 deep quarters field.

Multi-Agent Run Blocking Classification and Offensive Line Leverage
Rushing efficiency is heavily mediated by blocking structure. Evaluating rushing attempts purely through realized rushing yards conflates individual running back skill with offensive line push and run concept design. The engine utilizes an 11-player spatial-temporal transformer to classify the offensive run concept and quantify individual offensive line blocking matchups.

The network processes the dynamic coordination of the five offensive linemen (Center, Left/Right Guards, Left/Right Tackles) plus tight ends and fullbacks across the first 1.5 seconds post-snap. Run concepts are classified into seven discrete schemes: Duo, Outside Zone (Stretch), Inside Zone, Pin-Pull, Counter (Gap/Power), Trap, and Split Zone.

Classification relies on offensive line collective displacement vectors $\Delta P_{OL} \in \mathbb{R}^{5\times 2}$ and velocity divergence. In Outside Zone, all five linemen execute simultaneous, unilateral lateral bucket steps characterized by high lateral velocity (v_y >2.5 m/s) and uniform directional alignment ($\theta_{dir} \approx \pm 90^\circ$).

In Duo, the offensive line generates minimal lateral flow, executing vertical double-teams at the point of attack with guards and tackles driving interior defensive tackles backward ($\hat{v}_x$ >1.5 m/s) before peeling off to second-level linebackers. In Pin-Pull and Counter schemes, the architecture detects pulling linemen via characteristic inward-transverse trajectory loops where a backside guard or tackle pulls behind the center:

I
pull

(i)=I(
t∈[t
snap

,t+1.5]
max

|v
y,i

(t)|>3.0 m/s∧Δx
i

(t)<0)
Simultaneously, the engine models individual blocking assignments as a bipartite matching between offensive blockers and front-seven defensive players. Blocker-defender engagement is defined via continuous spatial proximity and opposing momentum vectors:

Engagement
ij

(t)=I(// p
off
i

(t)−p
def
j

(t) //
2

<1.2 yards∧
v

off

i

(t) ·
v

def
j

(t)<−0.5 //
v

off
i

// //
v

def
j

//)
From this engagement, the engine extracts the Blocker Leverage Tensor B
ij

(t), capturing pad level, driving angle, and boundary lane creation:

B
ij

(t)=[// p
off
i

(t)−p
def
j

(t) //
2

,Δ
v

ij

(t) ·
n
^

gap

,(p
carrier

(t)−p
def
j

(t)) ·
v

carrier

(t),Spread
gap

(t)]
T

By mapping blocking matchups, the engine isolates blocker push from running back vision. Conditioning on the classified run concept, defensive front alignment, and spatial gap widths at the point of handoff, the engine calculates Rushing Yards Over Expected (RYOE):

RYOE=Y
actual

−E[Y∣RunScheme,p
carrier

(t
handoff

),v
carrier

(t
handoff

),{B
ij

(t
handoff

)}
i=1
5

]
This metric isolates elite running back contact balance, burst, and cut-back decision-making from offensive line run-blocking quality.

Adaptive Conformal Inference and Fail-Closed Small-Sample Calibration
Uncalibrated quantile predictions generated by deep neural networks or gradient-boosted trees exhibit severe empirical under-coverage in finite samples. As proven by Lin, Trivedi, and Sun (2021), linear and non-linear quantile regressors suffer from an inherent under-coverage bias proportional to parameter dimensionality over sample size:

Coverage(α)≈α−(α−
2
1

)

n
d

Raw quantile outputs cannot be deployed directly into market trading engines without violating strict calibration bounds. To establish distribution-free, finite-sample predictive guarantees, continuous numeric lines (e.g., player rushing yards, passing props, game totals) are processed through Conformalized Quantile Regression (CQR).

CQR fits two base quantile regressors,
q
^

α
lo

(x) and
q
^

α
hi

(x), at nominal coverage 1−α, setting α
lo

=α/2 and α
hi

=1−α/2. Evaluating these estimators on a strictly held-out calibration set I
2

={(X
i

,Y
i

)}
i=1
n

, non-conformity scores E
i

are computed as:

E
i

=max(
q
^

α
lo

(X
i

)−Y
i

,Y
i

−
q
^

α
hi

(X
i

))
The non-conformity score E
i

represents the signed error magnitude by which the true response falls outside the predicted quantile corridor. The conformal inflation adjustment Q
1−α

(E;I
2

) is evaluated as the k-th smallest value of {E
i

}
i=1
n

, where the rank index k is calculated as:

k=⌈(n+1)(1−α)⌉
For any new test instance X
n+1

, the valid conformal prediction interval is constructed as:

C(X
n+1

)=[
q
^

α
lo

(X
n+1

)−Q
1−α

(E;I
2

),
q
^

α
hi

(X
n+1

)+Q
1−α

(E;I
2

)]
Under the assumption of data exchangeability, the marginal coverage guarantee holds exactly in finite samples:

P(Y
n+1

∈C(X
n+1

))≥1−α
The operational sequence governing conformal inference execution proceeds through distinct stages:

First, base models output lower and upper quantiles
q
^

α
lo

,
q
^

α
hi

via pinball loss optimization. Second, calibration residuals are evaluated against empirical labels to generate non-conformity scores E
i

. Third, the conformal rank index k=⌈(n+1)(1−α)⌉ is computed against calibration set size n. If k≤n, the empirical quantile Q
1−α

=E
(k)

is extracted and applied symmetrically to construct valid prediction intervals. Conversely, if k>n, the sample size is mathematically insufficient to certify the nominal coverage level, immediately triggering the fail-closed operational policy.

A critical vulnerability occurs in scarce calibration environments, specifically concerning the handling of small sample sizes where n<⌈1/α⌉−1. For a nominal coverage level of 90% (α=0.10), the formula dictates that for n=5:

k=⌈(5+1)(1−0.10)⌉=⌈6×0.90⌉=⌈5.4⌉=6
Because k=6>n=5, the required rank index strictly exceeds the sample size. Mathematically, the empirical cumulative distribution function cannot exceed 1.0; evaluating the quantile at (1−α)(1+1/n)=0.90×1.20=1.08>1.0 forces the empirical quantile to +∞.

In naive production software, engineers frequently implement a heuristic clamp, forcing k=min(n,⌈(n+1)(1−α)⌉)=5. The mathematical audit of the Beexly/Sports codebase demonstrates that this clamping practice is catastrophic. Under data exchangeability, selecting the maximum calibration residual E
(n)

at n=5 yields a true finite-sample coverage of:

$$P(Y_{n+1} \in C(X_{n+1})) = \frac{n+1}{n} = \frac{6}{5} \approx 83.33\%$$

Quantile clamping falsely presents an interval as possessing 90% confidence while delivering a 6.67% under-coverage defect. This generates deceptive "fake tightness," leading downstream automated execution gates to misprice risk.

The engine enforces a strict fail-closed operational policy: all quantile clamping is banned in cqr.ts. When $n < \lceil 1/\alpha \rceil - 1$, the engine either emits an infinite interval $[-\infty, +\infty]$ or abstains entirely by triggering an automated "No-Bet" execution block.

In scarce early-season segments where data splitting is too costly, the engine substitutes split CQR with Jackknife+. Utilizing leave-one-out fitted models $\hat{\mu}_{-i}$ and residuals $R_i^{LOO} = |Y_i - \hat{\mu}_{-i}(X_i)|$

−i

(X
i

)∣, Jackknife+ constructs intervals without a split calibration set while providing a distribution-free coverage guarantee of at least 1−2α:

C
jackknife+

(X
n+1

)=[q
α
−

(
μ
^

−i

(X
n+1

)−R
i
LOO

),q
1−α
+

(
μ
^

−i

(X

n+1

)+R
i
LOO

)]
Applying continuous CQR residual math to binary moneyline outcomes ($Y \in \{0,1\}$) is mathematically invalid. As proven by Barber et al. (2020), distribution-free confidence intervals for binary conditional probabilities $\pi(X)=P(Y=1 \mid X)$ of bounded length are impossible without structural assumptions. Continuous residual expansion cannot output a calibrated probability distribution.

For binary side markets, the engine deploys Generalized Venn-Abers calibration. Venn-Abers non-parametrically computes exact multiprobability bounds [p
0

,p
1

] by fitting dual monotonic isotonic regressions via the Pool Adjacent Violators Algorithm (PAVA) to the calibration set augmented with hypothetical test labels (X
n+1

,0) and (X
n+1

,1):

p
0

=
P
^
(Y
n+1

=1 | I
2

∪{(X
n+1

,0)}),p
1

=
P
^
(Y
n+1

=1∣I
2

∪{(X
n+1

,1)})
The bound width Δp=p
1

−p
0

explicitly isolates and quantifies epistemic uncertainty stemming from small sample sizes. Tied raw scores are pre-coalesced prior to PAVA execution to prevent non-unique knot assignments. The engine's selective execution gate consumes these bounds directly: if Δp>0.20, the model's epistemic doubt is deemed excessive, and the system issues a fail-closed pass.

Over a 272-game NFL regular season, strict exchangeability is violated by weather transitions, roster churn, and tactical counter-adjustments. To counter non-exchangeable distribution drift, the system integrates Adaptive Conformal Inference (ACI) and Rolling-Origin Conformal Prediction. ACI dynamically adjusts the nominal miscoverage level α
t

after every observed outcome based on empirical step error:

α
t+1

=α
t

+γ(α−err
t

),where err
t

=I(Y
t

∈
/
C
t

(X
t

))
where γ>0 governs adaptation responsiveness. Simultaneously, Rolling-Origin Conformal Prediction restricts calibration to a sliding window of the m=32–64 most recent completed games, maintaining local stationarity and preventing historical early-season dynamics from biasing late-season playoff projections.

Calibration Strategy	Target Market Domain	Exchangeability Assumption	Small-Sample Policy (n<9,α=0.10)	Production Execution Gate
Split CQR (Standard)	Continuous lines (player props, totals).	Assumes exchangeable residual distribution.	Strictly fail-closed: emits [−∞,+∞].	Automated "No-Bet" execution block.
Jackknife+	Scarce continuous lines (early-season, rookie props).	Assumes symmetric training point exchangeability.	Retains 1−2α validity without data-splitting loss.	Publishable bounds if interval width ≤ threshold.
Venn-Abers Calibration	Binary sides (moneylines, 2-way outcomes).	Assumes label/feature exchangeability.		

Emits exact multiprobability bounds [p
0

,p
1

].

Fail-closed if epistemic spread p
1

−p
0

>0.20.

Adaptive Conformal (ACI)
In-season sequential lines under regime drift.

Non-exchangeable; adapts to arbitrary shift.

Requires burn-in history; adapts nominal quantile α
t

.

Active tracking of empirical coverage error.

Rolling-Origin CP
Dynamic team strength margins, closing lines.

Locally stationary over sliding window m.

Fails closed if window m<⌈1/α⌉.

Gated by rolling calibration set size m≥32.

Low-Latency Streaming Pipeline and Shadow System Topology
To execute real-time spatial inference on 10Hz tracking data without violating live market execution thresholds, the ingestion and modeling architecture must achieve end-to-end processing latencies under 20 milliseconds per frame. The operational data pipeline is

constructed around the Apache Arrow memory format, zero-copy Inter-Process Communication (IPC), and GPU-accelerated inference pipelines.

Incoming tracking streams from stadium RFID receiver grids are ingested via Apache Arrow Flight, a high-throughput RPC framework optimized for columnar data transfers that eliminates CPU serialization bottlenecks. Prior to model processing, incoming tracking matrices pass through a strict state-transition verification gate modeled on the VERSA framework.

The audit reveals that nearly 19% of raw sports event logs suffer from spatial or temporal inconsistencies (e.g., impossible player teleportation velocities // v

// >13 m/s, mismatched coordinate axes, or reversed play directions). VERSA acts as a fail-closed filter: frames exhibiting corrupt state-transition kinematics are purged or spline-interpolated before reaching GPU memory.

Once verified, Arrow record batches are mapped directly into GPU memory via CUDA IPC using RAPIDS cuDF, bypassing host-to-device memory copies. Deep learning inference is orchestrated via the NVIDIA Triton Inference Server utilizing TensorRT-optimized execution graphs. Models are organized into an ensemble pipeline executed in a single forward pass: the factorized coverage transformer decodes defensive assignments; the non-autoregressive PlayGen-MoG module forecasts multi-agent continuation trajectories; and the Yurko CTMDP regression head evaluates within-play value generation.

The extracted spatial tracking metrics are routed directly into the Beexly/Sports persistent shadow evaluation pipeline (shadow-evaluation-pass.ts). The system decouples pure athletic performance modeling from market pricing dynamics to eliminate target leakage.

Incorporating betting market closing lines as input features into team or player performance regressors induces circular confirmation bias: the model learns to parrot the consensus oddsmaker rather than discovering structural athletic edges. Therefore, performance tracking metrics (ESOE, RYOE, CPOE, defensive disguise rates) feed solely into the TeamStrengthFilter—an Ensemble Transform Kalman Filter (ETKF) that models latent team ability as a sequential Monte Carlo particle filter:

θ
t

=Fθ
t−1

+w
t

,w
t

~N(0,Q)
The ETKF state survives serverless restarts through PostgreSQL persistence using FilterStateSnapshot and ShadowSignal entities, ensuring that team strength memory remains consistent across operational restarts.

Market pricing dynamics are tracked independently by a Hawkes self-exciting point process (hawkes-steam.ts). Betting line movements and order arrivals are modeled as jump processes where the background intensity μ captures organic liquidity, while self-exciting cross-excitation parameters α
ij

e
−β(t−t
k

)
detect clustered, informed sharp betting pressure ("steam"):

λ
market

(t)=μ+
t
k

<t
∑

αe
−β(t−t
k

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The LiveOrchestrator reconciles model predictions against the Hawkes-filtered market consensus. Live capital deployment is gated through an anytime-valid sequential Forecast-Skill E-process (forecast-skill-eprocess.ts), rooted in Ville's martingale inequality. The E-process evaluates the running likelihood ratio martingale M
t

testing the null hypothesis H

0

that the model's calibrated probabilities P
model

possess no predictive edge over the de-vigged market implied probabilities P
market

:

M
t

=
k=1
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P
market

(Y
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|X
k

)
P
model

(Y
k

|X
k

)

,with E
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[M
t

]≤1
Under Ville's inequality, the probability that the martingale wealth M
t

ever breaches a threshold 1/α
threshold

at any stopping time τ is bounded:

P
H
0

(
t≥1
sup

M
t

≥
α
threshold

1

)≤α
threshold

Continuous monitoring of M
t

provides mathematically valid proof of forecast skill without inflating Type I error rates or falling victim to p-hacking. Only when M
t

multiplies capital beyond conservative thresholds does the engine unlock execution, passing the calibrated expected value through a robust fractional Kelly optimizer (robust-kelly.ts) to size deployment against market vigorish.

The end-to-end data transformation pipeline enforces strict latency and security constraints across all five processing tiers:

Ingestion Tier: Ingests raw 10Hz RFID coordinate chirps from stadium sensors via AWS Kinesis and S3 into Apache Arrow Flight streaming buffers.

Verification Tier: Executes the fail-closed VERSA verification gate, auditing frame sequence continuity, velocity bounds ($\|v\|$ ≤13 m/s), and spatial coordinate orientation.

Transport Tier: Transfers verified Arrow RecordBatches directly to GPU shared memory using CUDA IPC and RAPIDS cuDF, achieving zero CPU-to-GPU copy overhead.

Inference Tier: NVIDIA Triton Server executes TensorRT-compiled factorized attention models, generating frame-by-frame coverage attribution, PlayGen-MoG trajectory branches, and Yurko CTMDP valuations.

Execution Tier: LiveOrchestrator computes fail-closed CQR prediction intervals and Venn-Abers bounds, verifies that epistemic uncertainty $\Delta p \le 0.20$, confirms forecast-skill martingale growth ($M_t > \tau_{skill}$), and outputs executable orders sized via robust Kelly criteria.

Systematic Synthesis of Spatial Modeling Architectures
Integrating 10Hz micro-movement spatial tracking data into the Beexly/Sports engine establishes a continuous, physically grounded intelligence layer that supersedes discrete play-by-play modeling. By enforcing strict mathematical boundaries—eliminating target leakage through model decoupling, eradicating "fake tightness" via fail-closed conformal prediction, and formalizing spatial gravity via potential flow theory—the engine secures an institutional, un-arbitraged predictive edge across live NFL derivative and primary markets.

Pipeline Dimension	Baseline Play-by-Play Engine	Intermediate Spatial Engine	Institutional Micro-Movement Pipeline (2026 Target)
Data Ingestion Frequency			

Discrete event-level (1 row per play).

10Hz raw coordinate feeds (x,y only).

10Hz full kinematic tensor: (x,y,
v

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Data Cleaning & Integrity
Post-game manual box-score correction.

Rule-based velocity threshold filters (v<12 m/s).

Fail-closed VERSA state-transition verification; zero-copy Apache Arrow Flight IPC.

Agent Representation
Tabular feature vectors; role-dependent column indexing.

Spatio-temporal graph convolution (ST-GCN).

Factorized Spatio-Temporal Attention + Permutation-Equivariant Set Attention Blocks (SABs).

Trajectory Simulation
Autoregressive sequential LSTM; accumulates compounding drift.

Conditional Variational Autoencoder (CVAE).

Non-Autoregressive PlayGen-MoG with globally coupled mixture weights across all 22 agents.

Separation Metric
Gross Average Separation at pass arrival.

Separation distance relative to nearest defender.

Expected Separation Over Expected (ESOE) conditioned on B-spline curvature, break mechanics, and leverage tensors.

Coverage Attribution
Nearest defender heuristic at ball arrival (25% error).

Logistic regression on pre-snap player depths.

Frame-by-frame factorized transformer coverage ID (20 classes) + Safety Rotation Disguise Entropy (D
KL

).

Within-Play Valuation
Discrete EP / EPA at play start and end.

Regression-based yards-gained expected point lookup.

Continuous-time semi-martingale CTMDP + Yurko conditional density estimation + Potential Field Navier-Stokes pitch control.

Uncertainty Quantification
Uncalibrated quantile regression (exhibits systemic under-coverage).

Split CQR with heuristic quantile clamping (creates "fake tightness").

Fail-Closed CQR (bans clamping; enforces No-Bet when $n<\lceil 1/\alpha \rceil - 1$) + Generalized Venn-Abers bounds [p
0

,p
1

].

Distribution Drift Adaptation
Static seasonal retraining.

Ad-hoc recency weighting.

Adaptive Conformal Inference (ACI) + Rolling-Origin Conformal Prediction (m=32–64 games).

Execution Gating & Latency
Static p-value tests; >5s post-play computation.

Un-gated edge thresholds (EV>0.05); 500ms latency.
Triton GPU inference (<15ms) + Ville's Martingale E-Process (M
t

) + Hawkes Steam Detection + Robust Kelly.

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image.png Hermes I found real omissions-not just cleanup. Most important: My conformal fix was incomplete. cqr.ts contains the same small-sample quantile clamp, and its tests explicitly expect the ...
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